

MASTER BLUEPRINT v3.0 – PART 9A

COMPLETE POSTGRESQL / SUPABASE DATABASE SCHEMA

AUTHENTICATION, USERS, RBAC, TENANTS, SESSIONS & SECURITY

Version: 3.0

Status: Database Implementation Specification

301. DATABASE STANDARDS

Database:

PostgreSQL 17+

Platform:

Supabase

Primary Key:

UUID

UUID Generation:

gen_random_uuid()

Timezone:

UTC

Soft Delete:

deleted_at

Audit Columns:

Mandatory

STANDARD AUDIT COLUMNS

Every Business Table Contains:

id UUID PRIMARY KEY

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

created_by UUID

updated_by UUID

deleted_at TIMESTAMPTZ

302. ENUMS

user_status_enum

active

inactive

suspended

blocked

pending_verification

deleted

verification_status_enum

pending

submitted

approved

rejected

expired

tenant_type_enum

platform

agency

builder

enterprise

franchise

device_type_enum

desktop

mobile

tablet

api

session_status_enum

active

expired

revoked

logged_out

303. TENANTS TABLE

Purpose:

Multi Tenant Foundation

Table:

tenants

Columns:

id UUID PK

name VARCHAR(255)

slug VARCHAR(255) UNIQUE

type tenant_type_enum

logo_url TEXT

website_url TEXT

contact_email VARCHAR(255)

contact_phone VARCHAR(50)

status BOOLEAN DEFAULT TRUE

settings JSONB DEFAULT '{}'

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

created_by UUID

updated_by UUID

deleted_at TIMESTAMPTZ

Indexes:

slug

type

status

304. ROLES TABLE

Table:

roles

Columns:

id UUID PK

name VARCHAR(100)

slug VARCHAR(100)

description TEXT

is_system BOOLEAN

priority INTEGER

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

deleted_at TIMESTAMPTZ

Default Roles:

buyer_seller

broker

builder

agency

admin

super_admin

Indexes:

slug UNIQUE

priority

305. PERMISSIONS TABLE

Table:

permissions

Columns:

id UUID PK

module VARCHAR(100)

action VARCHAR(100)

permission_key VARCHAR(255)

description TEXT

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

Examples:

property.create

property.update

property.delete

lead.assign

lead.transfer

payment.refund

settings.manage

Indexes:

permission_key UNIQUE

module

306. ROLE_PERMISSIONS TABLE

Table:

role_permissions

Columns:

id UUID PK

role_id UUID FK

permission_id UUID FK

created_at TIMESTAMPTZ

Unique:

(role_id, permission_id)

307. USERS TABLE

Table:

users

Columns:

id UUID PK

tenant_id UUID FK

primary_role_id UUID FK

first_name VARCHAR(150)

last_name VARCHAR(150)

display_name VARCHAR(255)

email CITEXT

mobile VARCHAR(20)

avatar_url TEXT

gender VARCHAR(30)

date_of_birth DATE

city VARCHAR(150)

state VARCHAR(150)

country VARCHAR(150)

language VARCHAR(20)

timezone VARCHAR(100)

status user_status_enum

email_verified BOOLEAN
mobile_verified BOOLEAN
last_login_at TIMESTAMPTZ
created_at TIMESTAMPTZ
updated_at TIMESTAMPTZ
created_by UUID
updated_by UUID
deleted_at TIMESTAMPTZ

Indexes:

email UNIQUE
mobile UNIQUE
tenant_id
status
city
state

308. USER_ROLES TABLE

Purpose:

One User Multiple Roles Support

Table:

user_roles

Columns:

id UUID PK

user_id UUID FK

role_id UUID FK

assigned_at TIMESTAMPTZ

assigned_by UUID

Unique:

(user_id, role_id)

309. USER_PROFILES TABLE

Table:

user_profiles

Columns:

id UUID PK

user_id UUID FK

bio TEXT

website_url TEXT

linkedin_url TEXT

facebook_url TEXT

instagram_url TEXT

twitter_url TEXT

youtube_url TEXT

profile_completion INTEGER

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

Indexes:

user_id UNIQUE

310. USER_PREFERENCES TABLE

Table:

user_preferences

Columns:

id UUID PK

user_id UUID FK

language VARCHAR(20)

currency VARCHAR(20)

theme VARCHAR(20)
email_notifications BOOLEAN
sms_notifications BOOLEAN
push_notifications BOOLEAN
whatsapp_notifications BOOLEAN
created_at TIMESTAMPTZ
updated_at TIMESTAMPTZ
Indexes:
user_id UNIQUE

311. USER_SESSIONS TABLE

Purpose:

Login Tracking

Columns:

id UUID PK
user_id UUID FK
device_id UUID FK
session_token TEXT
refresh_token TEXT
ip_address INET
user_agent TEXT
country VARCHAR(100)
city VARCHAR(100)
login_at TIMESTAMPTZ
expires_at TIMESTAMPTZ
status session_status_enum
logout_at TIMESTAMPTZ

Indexes:

user_id

status

login_at

312. USER_DEVICES TABLE

Columns:

id UUID PK

user_id UUID FK

device_name VARCHAR(255)

device_type device_type_enum

device_identifier TEXT

browser VARCHAR(255)

os VARCHAR(255)

push_token TEXT

is_trusted BOOLEAN

last_used_at TIMESTAMPTZ

created_at TIMESTAMPTZ

Indexes:

user_id

device_type

313. OTP_REQUESTS TABLE

Purpose:

OTP Verification

Columns:

id UUID PK

user_id UUID FK

mobile VARCHAR(20)

email VARCHAR(255)
otp_hash TEXT
otp_type VARCHAR(50)
attempts INTEGER
expires_at TIMESTAMPTZ
verified_at TIMESTAMPTZ
ip_address INET
created_at TIMESTAMPTZ
Indexes:
mobile
email
expires_at

314. LOGIN_ATTEMPTS TABLE

Purpose:
Brute Force Protection
Columns:
id UUID PK
email VARCHAR(255)
mobile VARCHAR(20)
ip_address INET
success BOOLEAN
failure_reason TEXT
attempted_at TIMESTAMPTZ
Indexes:
ip_address
attempted_at
success

315. BLOCKED_IPS TABLE

Columns:

id UUID PK

ip_address INET

reason TEXT

blocked_until TIMESTAMPTZ

created_at TIMESTAMPTZ

Indexes:

ip_address UNIQUE

316. SECURITY_LOGS TABLE

Columns:

id UUID PK

user_id UUID FK

event_type VARCHAR(100)

severity VARCHAR(50)

description TEXT

ip_address INET

metadata JSONB

created_at TIMESTAMPTZ

Indexes:

user_id

event_type

severity

created_at

317. AUDIT_LOGS TABLE

Purpose:

Immutable Tracking

Columns:

id UUID PK

user_id UUID FK

module VARCHAR(100)

entity_type VARCHAR(100)

entity_id UUID

action VARCHAR(100)

old_values JSONB

new_values JSONB

ip_address INET

user_agent TEXT

created_at TIMESTAMPTZ

Indexes:

user_id

module

entity_type

entity_id

created_at

318. API_KEYS TABLE

Future Enterprise API

Columns:

id UUID PK

tenant_id UUID FK

name VARCHAR(255)

api_key_hash TEXT

status BOOLEAN

rate_limit INTEGER

expires_at TIMESTAMPTZ

created_at TIMESTAMPTZ

Indexes:

tenant_id

319. FEATURE_FLAGS TABLE

Columns:

id UUID PK

flag_key VARCHAR(255)

name VARCHAR(255)

description TEXT

enabled BOOLEAN

config JSONB

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

Indexes:

flag_key UNIQUE

320. PLATFORM_SETTINGS TABLE

Columns:

id UUID PK

setting_key VARCHAR(255)

setting_value JSONB

description TEXT

updated_at TIMESTAMPTZ

Indexes:

setting_key UNIQUE

321. RLS POLICIES

Users:

Can Read Own Profile

Can Update Own Profile

Cannot Read Other Users

Broker:

Can Read Assigned Data

Cannot Read Other Broker Data

Agency:

Can Read Agency Users

Can Read Agency Leads

Admin:

Moderation Access

Limited Tenant Access

Super Admin:

Full Access

322. TRIGGERS

updated_at trigger

audit trigger

soft delete trigger

activity trigger

security_trigger

323. PART 9A TABLE COUNT

tenants

roles

permissions

role_permissions

users

user_roles

user_profiles

user_preferences

user_sessions

user_devices

otp_requests

login_attempts

blocked_ips

security_logs

audit_logs

api_keys

feature_flags

platform_settings

Total Tables:

18

PART 9A COMPLETE

MASTER BLUEPRINT v3.0 – PART 9B

COMPLETE PROPERTY DATABASE SCHEMA

Version: 3.0

Status: Database Implementation Specification

324. PROPERTY ENUMS

property_type_enum

apartment

villa

bungalow

penthouse

row_house

farm_house

plot

commercial_office

shop

showroom

warehouse

industrial

agricultural

pg

hostel

co_living

land

other

listing_type_enum

sale

rent

lease

resale

new_launch

property_status_enum

draft

pending_review

approved

rejected

active

sold

rented

expired

archived

hidden

furnishing_status_enum

unfurnished

semi_furnished

fully_furnished

ownership_type_enum

owner

broker

builder

agency

325. PROPERTIES TABLE

Master Property Table

Columns:

id UUID PK

tenant_id UUID FK

owner_user_id UUID FK

property_code VARCHAR(100)

title VARCHAR(500)

slug VARCHAR(500)

description TEXT

property_type property_type_enum

listing_type listing_type_enum

ownership_type ownership_type_enum

status property_status_enum

featured BOOLEAN

verified BOOLEAN

geo_verified BOOLEAN

rera_verified BOOLEAN

published_at TIMESTAMPTZ

expires_at TIMESTAMPTZ

meta_title VARCHAR(255)

meta_description TEXT

canonical_url TEXT

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

created_by UUID

updated_by UUID

deleted_at TIMESTAMPTZ

Indexes:

slug UNIQUE

property_type

listing_type

status

owner_user_id

tenant_id

featured

published_at

326. PROPERTY_DETAILS TABLE

Detailed Specifications

Columns:

id UUID PK

property_id UUID FK

bhk INTEGER

bedrooms INTEGER

bathrooms INTEGER

balconies INTEGER

servant_rooms INTEGER

floor_number INTEGER

total_floors INTEGER

property_age INTEGER

furnishing furnishing_status_enum

facing VARCHAR(50)

corner_property BOOLEAN

gated_community BOOLEAN

ready_to_move BOOLEAN

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

Indexes:

property_id UNIQUE

bhk

furnishing

327. PROPERTY_AREA_DETAILS TABLE

Columns:

id UUID PK

property_id UUID FK

super_builtup_area NUMERIC(12,2)

carpet_area NUMERIC(12,2)

builtup_area NUMERIC(12,2)

plot_area NUMERIC(12,2)

area_unit VARCHAR(30)

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

Indexes:

property_id UNIQUE

328. PROPERTY_PRICING TABLE

Columns:

id UUID PK

property_id UUID FK

price NUMERIC(15,2)

rent_amount NUMERIC(15,2)

deposit_amount NUMERIC(15,2)

maintenance_amount NUMERIC(15,2)

booking_amount NUMERIC(15,2)

price_per_sqft NUMERIC(12,2)

negotiable BOOLEAN

currency VARCHAR(10)

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

Indexes:

property_id UNIQUE

price

rent_amount

329. PROPERTY_LOCATION TABLE

Columns:

id UUID PK

property_id UUID FK

country VARCHAR(100)

state VARCHAR(100)

district VARCHAR(100)

city VARCHAR(100)

locality VARCHAR(255)

society VARCHAR(255)

address TEXT

pincode VARCHAR(20)

latitude NUMERIC(12,8)

longitude NUMERIC(12,8)

geography POINT

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

Indexes:

city

locality

pincode

latitude

longitude

PostGIS Geography Index

330. PROPERTY_AMENITIES MASTER

Table:

amenities

Columns:

id UUID PK

name VARCHAR(255)

slug VARCHAR(255)

icon VARCHAR(255)

category VARCHAR(100)

is_active BOOLEAN

created_at TIMESTAMPTZ

Indexes:

slug UNIQUE

category

331. PROPERTY_AMENITY_MAPPING

Table:

property_amenities

Columns:

id UUID PK

property_id UUID FK

amenity_id UUID FK
created_at TIMESTAMPTZ
Unique:
(property_id, amenity_id)

332. PROPERTY_MEDIA TABLE

Columns:

id UUID PK
property_id UUID FK
media_type VARCHAR(50)
file_url TEXT
thumbnail_url TEXT
file_size BIGINT
mime_type VARCHAR(100)
width INTEGER
height INTEGER
duration INTEGER
sort_order INTEGER
is_primary BOOLEAN
status BOOLEAN
created_at TIMESTAMPTZ

Indexes:

property_id
media_type
sort_order

333. PROPERTY_DOCUMENTS TABLE

Columns:

id UUID PK
property_id UUID FK
document_type VARCHAR(100)
title VARCHAR(255)
file_url TEXT
file_size BIGINT
verified BOOLEAN
uploaded_by UUID
created_at TIMESTAMPTZ
Indexes:
property_id
document_type

334. PROPERTY_TAGS TABLE

Columns:
id UUID PK
name VARCHAR(100)
slug VARCHAR(100)
Indexes:
slug UNIQUE

335. PROPERTY_TAG_MAPPING

Columns:
id UUID PK
property_id UUID FK
tag_id UUID FK
Unique:
(property_id, tag_id)

336. PROPERTY_VIEWS TABLE

Analytics

Columns:

id UUID PK

property_id UUID FK

user_id UUID FK NULL

ip_address INET

device VARCHAR(100)

viewed_at TIMESTAMPTZ

Indexes:

property_id

viewed_at

user_id

337. PROPERTY_FAVORITES TABLE

Columns:

id UUID PK

property_id UUID FK

user_id UUID FK

saved_at TIMESTAMPTZ

Unique:

(property_id, user_id)

Indexes:

user_id

property_id

338. PROPERTY_COMPARISONS TABLE

Columns:

id UUID PK

user_id UUID FK

property_id UUID FK

created_at TIMESTAMPTZ

Indexes:

user_id

property_id

339. PROPERTY_REPORTS TABLE

User Reported Listings

Columns:

id UUID PK

property_id UUID FK

reported_by UUID FK

reason VARCHAR(255)

description TEXT

status VARCHAR(50)

reviewed_by UUID

reviewed_at TIMESTAMPTZ

created_at TIMESTAMPTZ

Indexes:

property_id

status

340. PROPERTY_STATUS_LOGS TABLE

Columns:

id UUID PK

property_id UUID FK
old_status property_status_enum
new_status property_status_enum
changed_by UUID
remarks TEXT
created_at TIMESTAMPTZ

Indexes:

property_id
created_at

341. PROPERTY_PRICE_HISTORY TABLE

Columns:

id UUID PK
property_id UUID FK
old_price NUMERIC(15,2)
new_price NUMERIC(15,2)
changed_by UUID
created_at TIMESTAMPTZ

Indexes:

property_id
created_at

342. PROPERTY_CONTACT_LOGS TABLE

Tracks:

Call Click
WhatsApp Click
Lead Form

Columns:

id UUID PK
property_id UUID FK
user_id UUID FK
action_type VARCHAR(100)
ip_address INET
created_at TIMESTAMPTZ
Indexes:
property_id
action_type

343. PROPERTY_SEO TABLE

Columns:
id UUID PK
property_id UUID FK
meta_title VARCHAR(255)
meta_description TEXT
meta_keywords TEXT
schema_json JSONB
og_title VARCHAR(255)
og_description TEXT
created_at TIMESTAMPTZ
updated_at TIMESTAMPTZ
Indexes:
property_id UNIQUE

344. PROPERTY_GEO_VERIFICATION TABLE

Columns:
id UUID PK

property_id UUID FK
verified_by UUID
verification_method VARCHAR(100)
latitude NUMERIC(12,8)
longitude NUMERIC(12,8)
distance_meters NUMERIC(10,2)
verified_at TIMESTAMPTZ
Indexes:
property_id
verified_at

345. PROPERTY_ANALYTICS_DAILY TABLE

Columns:
id UUID PK
property_id UUID FK
date DATE
views INTEGER
unique_views INTEGER
favorites INTEGER
leads INTEGER
calls INTEGER
whatsapp_clicks INTEGER
visit_requests INTEGER
created_at TIMESTAMPTZ
Unique:
(property_id, date)

346. PROPERTY_SEARCH_INDEX TABLE

Precomputed Search Table

Columns:

id UUID PK

property_id UUID FK

search_text TSVECTOR

city VARCHAR(255)

locality VARCHAR(255)

property_type VARCHAR(100)

price NUMERIC(15,2)

status VARCHAR(100)

updated_at TIMESTAMPTZ

Indexes:

GIN(search_text)

city

property_type

price

347. PROPERTY_RECOMMENDATIONS TABLE

Columns:

id UUID PK

property_id UUID FK

recommended_property_id UUID FK

score NUMERIC(10,4)

algorithm VARCHAR(100)

created_at TIMESTAMPTZ

Indexes:

property_id

recommended_property_id

348. PROPERTY_DUPLICATE_DETECTION TABLE

Columns:

id UUID PK

property_id UUID FK

duplicate_property_id UUID FK

similarity_score NUMERIC(5,2)

status VARCHAR(50)

created_at TIMESTAMPTZ

Indexes:

property_id

duplicate_property_id

349. PROPERTY_AUDIT_TRAIL TABLE

Columns:

id UUID PK

property_id UUID FK

action VARCHAR(100)

old_data JSONB

new_data JSONB

performed_by UUID

created_at TIMESTAMPTZ

Indexes:

property_id

created_at

350. PROPERTY_RLS POLICIES

Buyer:

Read Approved Properties

Cannot Modify

Owner:

Read Own Properties

Edit Own Properties

Delete Own Draft Properties

Broker:

Manage Own Listings

Agency:

Manage Agency Listings

Admin:

Moderation Access

Super Admin:

Full Access

351. PROPERTY TRIGGERS

Create Search Index

Update Search Index

Price History Tracking

Status Change Tracking

Analytics Sync

Audit Trail Creation

Duplicate Detection Queue

352. PART 9B TABLE COUNT

properties

property_details

property_area_details

property_pricing

property_location

amenities

property_amenities

property_media

property_documents

property_tags

property_tag_mapping

property_views

property_favorites

property_comparisons

property_reports

property_status_logs

property_price_history

property_contact_logs

property_seo

property_geo_verification

property_analytics_daily

property_search_index

property_recommendations

property_duplicate_detection

property_audit_trail

Total Tables:

25

Running Total (Part 9A + 9B):

43 Tables

PART 9B COMPLETE

MASTER BLUEPRINT v3.0 – PART 9C

COMPLETE PROJECTS, BUILDERS, INVENTORY, TOWERS, UNITS & RERA DATABASE SCHEMA

Version: 3.0

Status: Database Implementation Specification

353. PROJECT ENUMS

project_status_enum

draft

pending_review

approved

rejected

active

completed

sold_out

expired

archived

project_type_enum

residential

commercial

mixed_use

industrial

plotted_development

villa_project

township

other

possession_status_enum

new_launch

under_construction

ready_to_move

completed

delivered

unit_status_enum

available

reserved

booked

sold

blocked

cancelled

354. BUILDERS TABLE

Master Builder Organization

Columns:

id UUID PK

tenant_id UUID FK

owner_user_id UUID FK

builder_code VARCHAR(100)

company_name VARCHAR(255)

slug VARCHAR(255)
description TEXT
logo_url TEXT
website_url TEXT
email VARCHAR(255)
mobile VARCHAR(30)
founded_year INTEGER
experience_years INTEGER
head_office_city VARCHAR(150)
head_office_state VARCHAR(150)
status BOOLEAN
verified BOOLEAN
featured BOOLEAN
created_at TIMESTAMPTZ
updated_at TIMESTAMPTZ
deleted_at TIMESTAMPTZ
Indexes:
slug UNIQUE
company_name
featured
verified

355. BUILDER_PROFILES TABLE

Columns:

id UUID PK
builder_id UUID FK
about_company TEXT
mission TEXT

vision TEXT

awards TEXT

total_projects INTEGER

completed_projects INTEGER

ongoing_projects INTEGER

upcoming_projects INTEGER

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

Indexes:

builder_id UNIQUE

356. BUILDER_CONTACTS TABLE

Columns:

id UUID PK

builder_id UUID FK

contact_name VARCHAR(255)

designation VARCHAR(255)

email VARCHAR(255)

mobile VARCHAR(30)

is_primary BOOLEAN

created_at TIMESTAMPTZ

Indexes:

builder_id

357. BUILDER_OFFICES TABLE

Columns:

id UUID PK

builder_id UUID FK

office_name VARCHAR(255)

address TEXT

city VARCHAR(150)

state VARCHAR(150)

country VARCHAR(150)

pincode VARCHAR(20)

latitude NUMERIC(12,8)

longitude NUMERIC(12,8)

created_at TIMESTAMPTZ

Indexes:

builder_id

city

358. PROJECTS TABLE

Master Project Table

Columns:

id UUID PK

tenant_id UUID FK

builder_id UUID FK

project_code VARCHAR(100)

project_name VARCHAR(255)

slug VARCHAR(255)

description TEXT

project_type project_type_enum

status project_status_enum

possession_status possession_status_enum

featured BOOLEAN

verified BOOLEAN

rera_verified BOOLEAN
launch_date DATE
completion_date DATE
possession_date DATE
meta_title VARCHAR(255)
meta_description TEXT
published_at TIMESTAMPTZ
created_at TIMESTAMPTZ
updated_at TIMESTAMPTZ
deleted_at TIMESTAMPTZ

Indexes:

slug UNIQUE
builder_id
status
project_type
featured

359. PROJECT_LOCATION TABLE

Columns:

id UUID PK
project_id UUID FK
country VARCHAR(100)
state VARCHAR(100)
city VARCHAR(150)
locality VARCHAR(255)
address TEXT
pincode VARCHAR(20)
latitude NUMERIC(12,8)

longitude NUMERIC(12,8)

geography POINT

created_at TIMESTAMPTZ

Indexes:

city

locality

PostGIS Geography Index

360. PROJECT_PHASES TABLE

Columns:

id UUID PK

project_id UUID FK

phase_name VARCHAR(255)

phase_number INTEGER

description TEXT

start_date DATE

end_date DATE

status VARCHAR(50)

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

Indexes:

project_id

phase_number

361. PROJECT_TOWERS TABLE

Columns:

id UUID PK

project_id UUID FK

phase_id UUID FK
tower_name VARCHAR(255)
tower_code VARCHAR(100)
total_floors INTEGER
total_units INTEGER
status VARCHAR(50)
launch_date DATE
possession_date DATE
created_at TIMESTAMPTZ
updated_at TIMESTAMPTZ
Indexes:
project_id
phase_id
tower_code

362. PROJECT_FLOORS TABLE

Columns:

id UUID PK
tower_id UUID FK
floor_number INTEGER
description TEXT
created_at TIMESTAMPTZ

Indexes:

tower_id
floor_number

363. PROJECT_UNITS TABLE

Master Inventory Unit

Columns:

id UUID PK

project_id UUID FK

tower_id UUID FK

floor_id UUID FK

unit_number VARCHAR(100)

unit_type VARCHAR(100)

bhk INTEGER

carpet_area NUMERIC(12,2)

builtup_area NUMERIC(12,2)

saleable_area NUMERIC(12,2)

status unit_status_enum

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

Indexes:

project_id

tower_id

status

unit_number

364. PROJECT_UNIT_PRICING TABLE

Columns:

id UUID PK

unit_id UUID FK

base_price NUMERIC(15,2)

agreement_value NUMERIC(15,2)

maintenance_cost NUMERIC(15,2)

parking_cost NUMERIC(15,2)

other_charges NUMERIC(15,2)
discount_amount NUMERIC(15,2)
final_price NUMERIC(15,2)
created_at TIMESTAMPTZ
updated_at TIMESTAMPTZ
Indexes:
unit_id UNIQUE
final_price

365. PROJECT_INVENTORY_SUMMARY TABLE

Columns:
id UUID PK
project_id UUID FK
available_units INTEGER
reserved_units INTEGER
booked_units INTEGER
sold_units INTEGER
blocked_units INTEGER
updated_at TIMESTAMPTZ
Indexes:
project_id UNIQUE

366. PROJECT_AMENITIES TABLE

Columns:
id UUID PK
project_id UUID FK
amenity_id UUID FK
created_at TIMESTAMPTZ

Unique:

(project_id, amenity_id)

367. PROJECT_MEDIA TABLE

Columns:

id UUID PK

project_id UUID FK

media_type VARCHAR(50)

file_url TEXT

thumbnail_url TEXT

sort_order INTEGER

is_primary BOOLEAN

created_at TIMESTAMPTZ

Indexes:

project_id

media_type

368. PROJECT_BROCHURES TABLE

Columns:

id UUID PK

project_id UUID FK

title VARCHAR(255)

file_url TEXT

version VARCHAR(50)

created_at TIMESTAMPTZ

Indexes:

project_id

369. PROJECT_FLOOR_PLANS TABLE

Columns:

id UUID PK

project_id UUID FK

unit_type VARCHAR(100)

file_url TEXT

description TEXT

created_at TIMESTAMPTZ

Indexes:

project_id

unit_type

370. PROJECT_CONSTRUCTION_UPDATES TABLE

Columns:

id UUID PK

project_id UUID FK

title VARCHAR(255)

description TEXT

completion_percentage NUMERIC(5,2)

update_date DATE

created_by UUID

created_at TIMESTAMPTZ

Indexes:

project_id

update_date

371. PROJECT_RERA_DETAILS TABLE

Columns:

id UUID PK
project_id UUID FK
rera_number VARCHAR(255)
registration_date DATE
expiry_date DATE
authority_name VARCHAR(255)
verification_status VARCHAR(50)
certificate_url TEXT
verified_at TIMESTAMPTZ
created_at TIMESTAMPTZ
Indexes:
project_id UNIQUE
rera_number UNIQUE

372. PROJECT_DEVELOPERS TABLE

Joint Venture Support

Columns:

id UUID PK
project_id UUID FK
builder_id UUID FK
developer_role VARCHAR(100)
created_at TIMESTAMPTZ

Unique:

(project_id, builder_id)

373. PROJECT_LEGAL_DOCUMENTS TABLE

Columns:

id UUID PK

project_id UUID FK
document_type VARCHAR(100)
title VARCHAR(255)
file_url TEXT
verified BOOLEAN
created_at TIMESTAMPTZ

Indexes:

project_id
document_type

374. PROJECT_SITE_VISITS TABLE

Columns:

id UUID PK
project_id UUID FK
visitor_user_id UUID FK
visit_date TIMESTAMPTZ
status VARCHAR(50)
notes TEXT
created_at TIMESTAMPTZ

Indexes:

project_id
visitor_user_id

375. PROJECT_FAVORITES TABLE

Columns:

id UUID PK
project_id UUID FK
user_id UUID FK

saved_at TIMESTAMPTZ

Unique:

(project_id, user_id)

Indexes:

user_id

project_id

376. PROJECT_VIEWS TABLE

Columns:

id UUID PK

project_id UUID FK

user_id UUID FK NULL

ip_address INET

viewed_at TIMESTAMPTZ

Indexes:

project_id

viewed_at

377. PROJECT_ANALYTICS_DAILY TABLE

Columns:

id UUID PK

project_id UUID FK

date DATE

views INTEGER

unique_views INTEGER

brochure_downloads INTEGER

site_visits INTEGER

leads INTEGER

favorites INTEGER

created_at TIMESTAMPTZ

Unique:

(project_id, date)

378. PROJECT_PRICE_HISTORY TABLE

Columns:

id UUID PK

unit_id UUID FK

old_price NUMERIC(15,2)

new_price NUMERIC(15,2)

changed_by UUID

created_at TIMESTAMPTZ

Indexes:

unit_id

created_at

379. BUILDER_KYC_DETAILS TABLE

Columns:

id UUID PK

builder_id UUID FK

pan_number VARCHAR(50)

gst_number VARCHAR(50)

company_registration_number VARCHAR(100)

verification_status VARCHAR(50)

verified_at TIMESTAMPTZ

created_at TIMESTAMPTZ

Indexes:

builder_id UNIQUE

gst_number

380. BUILDER_RATINGS TABLE

Columns:

id UUID PK

builder_id UUID FK

user_id UUID FK

rating NUMERIC(2,1)

review TEXT

created_at TIMESTAMPTZ

Indexes:

builder_id

rating

381. BUILDER_AWARDS TABLE

Columns:

id UUID PK

builder_id UUID FK

award_name VARCHAR(255)

award_year INTEGER

description TEXT

created_at TIMESTAMPTZ

Indexes:

builder_id

382. PROJECT_SEARCH_INDEX TABLE

Columns:

id UUID PK

project_id UUID FK

search_text TSVECTOR

city VARCHAR(150)

project_type VARCHAR(100)

builder_name VARCHAR(255)

updated_at TIMESTAMPTZ

Indexes:

GIN(search_text)

city

project_type

383. PROJECT_AUDIT_TRAIL TABLE

Columns:

id UUID PK

project_id UUID FK

action VARCHAR(100)

old_data JSONB

new_data JSONB

performed_by UUID

created_at TIMESTAMPTZ

Indexes:

project_id

created_at

384. PROJECT RLS POLICIES

Public:

View Approved Projects

Builder:

Manage Own Projects

Own Inventory

Own Updates

Agency:

View Assigned Projects

Admin:

Moderate Projects

Super Admin:

Full Access

385. PROJECT TRIGGERS

Inventory Sync Trigger

Price History Trigger

Construction Update Trigger

Search Index Trigger

Audit Trigger

Analytics Trigger

RERA Verification Trigger

386. PART 9C TABLE COUNT

builders

builder_profiles

builder_contacts

builder_offices
projects
project_location
project_phases
project_towers
project_floors
project_units
project_unit_pricing
project_inventory_summary
project_amenities
project_media
project_brochures
project_floor_plans
project_construction_updates
project_rera_details
project_developers
project_legal_documents
project_site_visits
project_favorites
project_views
project_analytics_daily
project_price_history
builder_kyc_details
builder_ratings
builder_awards
project_search_index
project_audit_trail

Total Tables:

30

Running Total:

Part 9A = 18 Tables

Part 9B = 25 Tables

Part 9C = 30 Tables

Grand Total So Far:

73 Tables

PART 9C COMPLETE

MASTER BLUEPRINT v3.0 – PART 9D

COMPLETE LEAD CRM, LEAD ROUTING, VISITS, MESSAGING & COMMUNICATION DATABASE SCHEMA

Version: 3.0

Status: Database Implementation Specification

387. LEAD ENUMS

lead_status_enum

new

contacted

interested

follow_up

visit_scheduled

visit_completed

negotiation

won

lost

spam

lead_source_enum

property_form

project_form

call

whatsapp

referral

organic_search

advertisement

manual

api

lead_priority_enum

low

medium

high

urgent

388. LEADS TABLE

Master Lead Table

Columns:

id UUID PK

tenant_id UUID FK

lead_number VARCHAR(100)

source lead_source_enum

status lead_status_enum

priority lead_priority_enum

property_id UUID FK NULL

project_id UUID FK NULL
assigned_to UUID FK
owner_user_id UUID FK
first_name VARCHAR(255)
last_name VARCHAR(255)
email VARCHAR(255)
mobile VARCHAR(30)
city VARCHAR(150)
budget NUMERIC(15,2)
requirement TEXT
notes TEXT
is_duplicate BOOLEAN
created_at TIMESTAMPTZ
updated_at TIMESTAMPTZ
deleted_at TIMESTAMPTZ

Indexes:

lead_number UNIQUE
assigned_to
status
source
property_id
project_id
mobile
email

389. LEAD_SOURCES TABLE

Columns:

id UUID PK

source_name VARCHAR(255)

description TEXT

is_active BOOLEAN

created_at TIMESTAMPTZ

Indexes:

source_name UNIQUE

390. LEAD_ASSIGNMENTS TABLE

Assignment History

Columns:

id UUID PK

lead_id UUID FK

assigned_from UUID FK

assigned_to UUID FK

reason TEXT

assigned_at TIMESTAMPTZ

Indexes:

lead_id

assigned_to

391. LEAD_NOTES TABLE

Columns:

id UUID PK

lead_id UUID FK

created_by UUID FK

note TEXT

created_at TIMESTAMPTZ

Indexes:

lead_id

created_by

392. LEAD_FOLLOWUPS TABLE

Columns:

id UUID PK

lead_id UUID FK

assigned_user_id UUID FK

followup_date TIMESTAMPTZ

followup_type VARCHAR(100)

remarks TEXT

status VARCHAR(50)

completed_at TIMESTAMPTZ

created_at TIMESTAMPTZ

Indexes:

lead_id

followup_date

assigned_user_id

393. LEAD_ACTIVITIES TABLE

CRM Activity Timeline

Columns:

id UUID PK

lead_id UUID FK

user_id UUID FK

activity_type VARCHAR(100)

description TEXT

metadata JSONB

created_at TIMESTAMPTZ

Indexes:

lead_id

activity_type

created_at

394. LEAD_STATUS_HISTORY TABLE

Columns:

id UUID PK

lead_id UUID FK

old_status lead_status_enum

new_status lead_status_enum

changed_by UUID FK

remarks TEXT

created_at TIMESTAMPTZ

Indexes:

lead_id

created_at

395. LEAD_TAGS TABLE

Columns:

id UUID PK

name VARCHAR(100)

slug VARCHAR(100)

color VARCHAR(20)

created_at TIMESTAMPTZ

Indexes:

slug UNIQUE

396. LEAD_TAG_MAPPING TABLE

Columns:

id UUID PK

lead_id UUID FK

tag_id UUID FK

created_at TIMESTAMPTZ

Unique:

(lead_id, tag_id)

397. LEAD_SCORING TABLE

Columns:

id UUID PK

lead_id UUID FK

score INTEGER

score_reason TEXT

calculated_at TIMESTAMPTZ

Indexes:

lead_id

score

398. LEAD_DUPLICATES TABLE

Columns:

id UUID PK

lead_id UUID FK

duplicate_lead_id UUID FK

similarity_score NUMERIC(5,2)

created_at TIMESTAMPTZ

Indexes:

lead_id

duplicate_lead_id

399. LEAD_IMPORTS TABLE

Bulk Lead Import

Columns:

id UUID PK

uploaded_by UUID FK

file_name VARCHAR(255)

total_records INTEGER

successful_records INTEGER

failed_records INTEGER

status VARCHAR(50)

created_at TIMESTAMPTZ

Indexes:

uploaded_by

400. LEAD_EXPORTS TABLE

Columns:

id UUID PK

requested_by UUID FK

filters JSONB

file_url TEXT

created_at TIMESTAMPTZ

Indexes:

requested_by

401. LEAD_PIPELINES TABLE

Columns:

id UUID PK

tenant_id UUID FK

pipeline_name VARCHAR(255)

description TEXT

is_default BOOLEAN

created_at TIMESTAMPTZ

Indexes:

tenant_id

402. LEAD_PIPELINE_STAGES TABLE

Columns:

id UUID PK

pipeline_id UUID FK

stage_name VARCHAR(255)

stage_order INTEGER

color VARCHAR(20)

created_at TIMESTAMPTZ

Indexes:

pipeline_id

stage_order

403. VISIT_REQUESTS TABLE

Columns:

id UUID PK

lead_id UUID FK NULL

property_id UUID FK NULL

project_id UUID FK NULL

requested_by UUID FK

assigned_to UUID FK

visit_date TIMESTAMPTZ

status VARCHAR(50)

notes TEXT

created_at TIMESTAMPTZ

Indexes:

visit_date

assigned_to

status

404. VISIT_FEEDBACK TABLE

Columns:

id UUID PK

visit_request_id UUID FK

rating INTEGER

feedback TEXT

created_by UUID FK

created_at TIMESTAMPTZ

Indexes:

visit_request_id

405. VISIT_STATUS_HISTORY TABLE

Columns:

id UUID PK

visit_request_id UUID FK

old_status VARCHAR(50)

new_status VARCHAR(50)

changed_by UUID FK

created_at TIMESTAMPTZ

Indexes:

visit_request_id

406. CONVERSATIONS TABLE

Master Chat

Columns:

id UUID PK

tenant_id UUID FK

conversation_type VARCHAR(50)

property_id UUID FK NULL

project_id UUID FK NULL

created_by UUID FK

last_message_at TIMESTAMPTZ

created_at TIMESTAMPTZ

Indexes:

property_id

project_id

last_message_at

407. CONVERSATION_PARTICIPANTS TABLE

Columns:

id UUID PK

conversation_id UUID FK

user_id UUID FK

joined_at TIMESTAMPTZ

is_muted BOOLEAN

is_archived BOOLEAN

last_read_message_id UUID

Indexes:

conversation_id

user_id

Unique:

(conversation_id, user_id)

408. MESSAGES TABLE

Columns:

id UUID PK

conversation_id UUID FK

sender_id UUID FK

message_type VARCHAR(50)

message_text TEXT

reply_to_message_id UUID NULL

is_edited BOOLEAN

edited_at TIMESTAMPTZ

created_at TIMESTAMPTZ

Indexes:

conversation_id

sender_id

created_at

409. MESSAGE_ATTACHMENTS TABLE

Columns:

id UUID PK

message_id UUID FK
file_url TEXT
file_name VARCHAR(255)
mime_type VARCHAR(100)
file_size BIGINT
created_at TIMESTAMPTZ

Indexes:

message_id

410. MESSAGE_READS TABLE

Columns:

id UUID PK
message_id UUID FK
user_id UUID FK
read_at TIMESTAMPTZ

Indexes:

message_id

user_id

Unique:

(message_id, user_id)

411. MESSAGE_REACTIONS TABLE

Columns:

id UUID PK
message_id UUID FK
user_id UUID FK
reaction VARCHAR(50)
created_at TIMESTAMPTZ

Indexes:

message_id

412. MESSAGE_DELETIONS TABLE

Columns:

id UUID PK

message_id UUID FK

deleted_by UUID FK

deleted_at TIMESTAMPTZ

Indexes:

message_id

413. CHAT_BLOCKS TABLE

Columns:

id UUID PK

blocked_by UUID FK

blocked_user UUID FK

reason TEXT

created_at TIMESTAMPTZ

Indexes:

blocked_by

blocked_user

414. CALL_LOGS TABLE

Columns:

id UUID PK

lead_id UUID FK NULL

property_id UUID FK NULL

caller_user_id UUID FK
receiver_user_id UUID FK
call_type VARCHAR(50)
duration_seconds INTEGER
status VARCHAR(50)
created_at TIMESTAMPTZ

Indexes:

caller_user_id
receiver_user_id

415. WHATSAPP_LOGS TABLE

Columns:

id UUID PK
lead_id UUID FK NULL
user_id UUID FK
template_name VARCHAR(255)
status VARCHAR(50)
sent_at TIMESTAMPTZ

Indexes:

user_id
status

416. EMAIL_LOGS TABLE

Columns:

id UUID PK
lead_id UUID FK NULL
user_id UUID FK
subject VARCHAR(255)

status VARCHAR(50)
sent_at TIMESTAMPTZ

Indexes:

user_id
status

417. CRM_AUTOMATIONS TABLE

Columns:

id UUID PK
tenant_id UUID FK
automation_name VARCHAR(255)
trigger_event VARCHAR(100)
action_type VARCHAR(100)
config JSONB
is_active BOOLEAN
created_at TIMESTAMPTZ

Indexes:

tenant_id

418. CRM_REMINDERS TABLE

Columns:

id UUID PK
lead_id UUID FK
assigned_user_id UUID FK
title VARCHAR(255)
description TEXT
reminder_at TIMESTAMPTZ
status VARCHAR(50)

created_at TIMESTAMPTZ

Indexes:

assigned_user_id

reminder_at

419. LEAD ROUTING RULES TABLE

Columns:

id UUID PK

tenant_id UUID FK

rule_name VARCHAR(255)

rule_type VARCHAR(100)

priority INTEGER

conditions JSONB

assigned_user UUID FK

is_active BOOLEAN

created_at TIMESTAMPTZ

Indexes:

tenant_id

priority

420. LEAD ROUTING HISTORY TABLE

Columns:

id UUID PK

lead_id UUID FK

rule_id UUID FK

assigned_to UUID FK

reason TEXT

created_at TIMESTAMPTZ

Indexes:

lead_id

rule_id

421. CRM ANALYTICS DAILY TABLE

Columns:

id UUID PK

tenant_id UUID FK

date DATE

new_leads INTEGER

won_leads INTEGER

lost_leads INTEGER

conversion_rate NUMERIC(5,2)

avg_response_time INTEGER

created_at TIMESTAMPTZ

Unique:

(tenant_id, date)

422. CRM AUDIT TRAIL TABLE

Columns:

id UUID PK

entity_type VARCHAR(100)

entity_id UUID

action VARCHAR(100)

old_data JSONB

new_data JSONB

performed_by UUID FK

created_at TIMESTAMPTZ

Indexes:

entity_type

entity_id

423. CRM RLS POLICIES

Buyer/Seller:

View Own Leads Only

Broker:

View Assigned Leads

Manage Assigned Leads

Agency:

View Agency Leads

Assign Leads

Transfer Leads

Builder:

View Project Leads

Manage Project Leads

Admin:

Moderation Access

Super Admin:

Full Access

424. CRM TRIGGERS

Lead Assignment Trigger
Lead Scoring Trigger
Lead Duplicate Detection Trigger
Visit Reminder Trigger
Message Notification Trigger
Pipeline Update Trigger
CRM Analytics Trigger
Audit Trigger

425. PART 9D TABLE COUNT

leads
lead_sources
lead_assignments
lead_notes
lead_followups
lead_activities
lead_status_history
lead_tags
lead_tag_mapping
lead_scoring
lead_duplicates
lead_imports
lead_exports
lead_pipelines
lead_pipeline_stages
visit_requests
visit_feedback
visit_status_history

conversations
conversation_participants
messages
message_attachments
message_reads
message_reactions
message_deletions
chat_blocks
call_logs
whatsapp_logs
email_logs
crm_automations
crm_reminders
lead_routing_rules
lead_routing_history
crm_analytics_daily
crm_audit_trail

Total Tables:

35

Running Total:

Part 9A = 18

Part 9B = 25

Part 9C = 30

Part 9D = 35

Grand Total:

108 Tables

PART 9D COMPLETE

MASTER BLUEPRINT v3.0 – PART 9E

COMPLETE SUBSCRIPTIONS, PAYMENTS, INVOICES, REFUNDS, ADVERTISEMENTS & REVENUE DATABASE SCHEMA

Version: 3.0

Status: Database Implementation Specification

426. MONETIZATION ENUMS

plan_type_enum

free

basic

premium

enterprise

custom

subscription_status_enum

active

pending

expired

cancelled

suspended

trial

payment_status_enum

pending

processing

success

failed

cancelled

refunded

disputed

payment_method_enum

upi

credit_card

debit_card

net_banking

wallet

emi

bank_transfer

invoice_status_enum

draft

generated

paid

cancelled

refunded

ad_status_enum

draft

pending_review

approved

active

paused

expired

rejected

427. PLANS TABLE

Master Subscription Plans

Columns:

id UUID PK

plan_code VARCHAR(100)

plan_name VARCHAR(255)

plan_type plan_type_enum

description TEXT

monthly_price NUMERIC(15,2)

quarterly_price NUMERIC(15,2)

yearly_price NUMERIC(15,2)

currency VARCHAR(10)

is_active BOOLEAN

sort_order INTEGER

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

Indexes:

plan_code UNIQUE

plan_type

428. PLAN_FEATURES TABLE

Columns:

id UUID PK

plan_id UUID FK

feature_key VARCHAR(255)

feature_value TEXT

created_at TIMESTAMPTZ

Indexes:

plan_id

feature_key

Examples:

listing_limit

featured_limit

image_limit

video_limit

lead_limit

agent_limit

ad_credit_limit

429. SUBSCRIPTIONS TABLE

Columns:

id UUID PK

tenant_id UUID FK

user_id UUID FK

plan_id UUID FK

subscription_number VARCHAR(100)

status subscription_status_enum

start_date DATE

end_date DATE

renewal_date DATE

auto_renew BOOLEAN

amount_paid NUMERIC(15,2)

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

Indexes:

subscription_number UNIQUE

user_id

plan_id

status

430. SUBSCRIPTION_HISTORY TABLE

Columns:

id UUID PK

subscription_id UUID FK

old_plan_id UUID FK

new_plan_id UUID FK

action VARCHAR(100)

performed_by UUID

created_at TIMESTAMPTZ

Indexes:

subscription_id

431. SUBSCRIPTION_USAGE TABLE

Columns:

id UUID PK

subscription_id UUID FK

feature_key VARCHAR(255)

used_value INTEGER

remaining_value INTEGER

updated_at TIMESTAMPTZ

Indexes:

subscription_id

feature_key

432. PLAN_COUPONS TABLE

Columns:

id UUID PK

coupon_code VARCHAR(100)

discount_type VARCHAR(50)

discount_value NUMERIC(15,2)

max_uses INTEGER

used_count INTEGER

valid_from DATE

valid_to DATE

is_active BOOLEAN

created_at TIMESTAMPTZ

Indexes:

coupon_code UNIQUE

433. COUPON_REDEMPTIONS TABLE

Columns:

id UUID PK

coupon_id UUID FK

user_id UUID FK

subscription_id UUID FK

discount_amount NUMERIC(15,2)

redeemed_at TIMESTAMPTZ

Indexes:

coupon_id

user_id

434. PAYMENTS TABLE

Master Payment Table

Columns:

id UUID PK

tenant_id UUID FK

user_id UUID FK

subscription_id UUID FK NULL

invoice_id UUID FK NULL

payment_number VARCHAR(100)

gateway VARCHAR(100)

gateway_payment_id VARCHAR(255)

gateway_order_id VARCHAR(255)

status payment_status_enum

payment_method payment_method_enum

amount NUMERIC(15,2)

currency VARCHAR(10)

tax_amount NUMERIC(15,2)

total_amount NUMERIC(15,2)

paid_at TIMESTAMPTZ

created_at TIMESTAMPTZ

Indexes:

payment_number UNIQUE

gateway_payment_id

status

user_id

435. PAYMENT_WEBHOOKS TABLE

Columns:

id UUID PK

provider VARCHAR(100)
event_type VARCHAR(255)
payload JSONB
signature TEXT
processed BOOLEAN
processed_at TIMESTAMPTZ
created_at TIMESTAMPTZ

Indexes:

provider
event_type
processed

436. PAYMENT_RECONCILIATION TABLE

Columns:

id UUID PK
payment_id UUID FK
gateway_reference VARCHAR(255)
system_amount NUMERIC(15,2)
gateway_amount NUMERIC(15,2)
difference_amount NUMERIC(15,2)
status VARCHAR(50)
created_at TIMESTAMPTZ

Indexes:

payment_id

437. INVOICES TABLE

Columns:

id UUID PK

tenant_id UUID FK
user_id UUID FK
invoice_number VARCHAR(100)
invoice_status invoice_status_enum
invoice_date DATE
due_date DATE
subtotal NUMERIC(15,2)
tax_amount NUMERIC(15,2)
discount_amount NUMERIC(15,2)
total_amount NUMERIC(15,2)
pdf_url TEXT
created_at TIMESTAMPTZ

Indexes:

invoice_number UNIQUE
user_id
invoice_status

438. INVOICE_ITEMS TABLE

Columns:

id UUID PK
invoice_id UUID FK
item_name VARCHAR(255)
description TEXT
quantity INTEGER
unit_price NUMERIC(15,2)
tax_rate NUMERIC(5,2)
total_amount NUMERIC(15,2)
created_at TIMESTAMPTZ

Indexes:

invoice_id

439. GST_DETAILS TABLE

Columns:

id UUID PK

user_id UUID FK

gst_number VARCHAR(50)

legal_name VARCHAR(255)

billing_address TEXT

city VARCHAR(150)

state VARCHAR(150)

country VARCHAR(100)

created_at TIMESTAMPTZ

Indexes:

gst_number UNIQUE

440. REFUNDS TABLE

Columns:

id UUID PK

payment_id UUID FK

refund_number VARCHAR(100)

refund_amount NUMERIC(15,2)

refund_reason TEXT

status VARCHAR(50)

gateway_refund_id VARCHAR(255)

processed_at TIMESTAMPTZ

created_at TIMESTAMPTZ

Indexes:

refund_number UNIQUE

payment_id

441. REFUND_HISTORY TABLE

Columns:

id UUID PK

refund_id UUID FK

old_status VARCHAR(50)

new_status VARCHAR(50)

changed_by UUID

created_at TIMESTAMPTZ

Indexes:

refund_id

442. TOKEN_BOOKINGS TABLE

Columns:

id UUID PK

property_id UUID FK NULL

project_id UUID FK NULL

user_id UUID FK

payment_id UUID FK

token_amount NUMERIC(15,2)

status VARCHAR(50)

booked_at TIMESTAMPTZ

Indexes:

user_id

property_id

project_id

443. AD_PACKAGES TABLE

Columns:

id UUID PK

package_name VARCHAR(255)

duration_days INTEGER

impression_limit INTEGER

click_limit INTEGER

priority_score INTEGER

price NUMERIC(15,2)

is_active BOOLEAN

created_at TIMESTAMPTZ

Indexes:

package_name

444. AD_CAMPAIGNS TABLE

Columns:

id UUID PK

tenant_id UUID FK

user_id UUID FK

package_id UUID FK

campaign_name VARCHAR(255)

budget NUMERIC(15,2)

start_date DATE

end_date DATE

status ad_status_enum

created_at TIMESTAMPTZ

Indexes:

user_id

status

445. ADVERTISEMENTS TABLE

Columns:

id UUID PK

campaign_id UUID FK

ad_type VARCHAR(100)

target_type VARCHAR(100)

target_id UUID

title VARCHAR(255)

description TEXT

banner_url TEXT

status ad_status_enum

created_at TIMESTAMPTZ

Indexes:

campaign_id

status

446. AD_TARGETING TABLE

Columns:

id UUID PK

advertisement_id UUID FK

cities JSONB

states JSONB

property_types JSONB

budget_range JSONB

created_at TIMESTAMPTZ

Indexes:

advertisement_id

447. AD_IMPRESSIONS TABLE

Columns:

id UUID PK

advertisement_id UUID FK

user_id UUID FK NULL

ip_address INET

impression_at TIMESTAMPTZ

Indexes:

advertisement_id

impression_at

448. AD_CLICKS TABLE

Columns:

id UUID PK

advertisement_id UUID FK

user_id UUID FK NULL

ip_address INET

clicked_at TIMESTAMPTZ

Indexes:

advertisement_id

clicked_at

449. FEATURED_LISTINGS TABLE

Columns:

id UUID PK
property_id UUID FK NULL
project_id UUID FK NULL
campaign_id UUID FK
priority_score INTEGER
start_date DATE
end_date DATE
created_at TIMESTAMPTZ

Indexes:

property_id

project_id

450. AD_LEADS TABLE

Columns:

id UUID PK
advertisement_id UUID FK
lead_id UUID FK
created_at TIMESTAMPTZ

Indexes:

advertisement_id

lead_id

451. REVENUE_ANALYTICS_DAILY TABLE

Columns:

id UUID PK
date DATE
subscription_revenue NUMERIC(15,2)
advertisement_revenue NUMERIC(15,2)

token_booking_revenue NUMERIC(15,2)

refund_amount NUMERIC(15,2)

net_revenue NUMERIC(15,2)

created_at TIMESTAMPTZ

Unique:

(date)

452. COMMISSIONS TABLE

Future Marketplace Support

Columns:

id UUID PK

payment_id UUID FK

commission_type VARCHAR(100)

commission_percentage NUMERIC(5,2)

commission_amount NUMERIC(15,2)

created_at TIMESTAMPTZ

Indexes:

payment_id

453. PAYOUTS TABLE

Future Partner Payouts

Columns:

id UUID PK

user_id UUID FK

amount NUMERIC(15,2)

status VARCHAR(50)

processed_at TIMESTAMPTZ

created_at TIMESTAMPTZ

Indexes:

user_id

status

454. REVENUE_AUDIT_TRAIL TABLE

Columns:

id UUID PK

entity_type VARCHAR(100)

entity_id UUID

action VARCHAR(100)

old_data JSONB

new_data JSONB

performed_by UUID

created_at TIMESTAMPTZ

Indexes:

entity_type

entity_id

455. MONETIZATION RLS POLICIES

Users:

Own Payments

Own Invoices

Own Subscriptions

Own Ads

Agency:

Agency Revenue Data

Agency Campaigns

Builder:

Own Project Campaigns

Admin:

Revenue Monitoring

Super Admin:

Full Access

456. MONETIZATION TRIGGERS

Subscription Activation Trigger

Invoice Generation Trigger

Payment Success Trigger

Refund Trigger

Revenue Analytics Trigger

Ad Impression Counter Trigger

Ad Click Counter Trigger

Coupon Redemption Trigger

Audit Trigger

457. PART 9E TABLE COUNT

plans

plan_features

subscriptions

subscription_history

subscription_usage

plan_coupons

coupon_redemptions
payments
payment_webhooks
payment_reconciliation
invoices
invoice_items
gst_details
refunds
refund_history
token_bookings
ad_packages
ad_campaigns
advertisements
ad_targeting
ad_impressions
ad_clicks
featured_listings
ad_leads
revenue_analytics_daily
commissions
payouts
revenue_audit_trail

Total Tables:

28

Running Total:

Part 9A = 18

Part 9B = 25

Part 9C = 30

Part 9D = 35

Part 9E = 28

Grand Total:

136 Tables

PART 9E COMPLETE

MASTER BLUEPRINT v3.0 – PART 9F

COMPLETE NOTIFICATIONS, ANALYTICS, CMS, SUPPORT, SECURITY, AUDIT, SYSTEM FUNCTIONS, VIEWS & ENTERPRISE DATABASE ARCHITECTURE

Version: 3.0

Status: Final Database Implementation Specification

458. NOTIFICATION ENUMS

notification_channel_enum

email

sms

whatsapp

push

in_app

notification_status_enum

queued

processing

sent

delivered

read

failed

cancelled

459. NOTIFICATIONS TABLE

Master Notification Table

Columns:

id UUID PK

tenant_id UUID FK

user_id UUID FK

channel notification_channel_enum

title VARCHAR(255)

message TEXT

payload JSONB

status notification_status_enum

sent_at TIMESTAMPTZ

read_at TIMESTAMPTZ

created_at TIMESTAMPTZ

Indexes:

user_id

status

channel

created_at

460. NOTIFICATION_TEMPLATES TABLE

Columns:

id UUID PK

template_key VARCHAR(255)

channel notification_channel_enum

name VARCHAR(255)

subject VARCHAR(255)

content TEXT

variables JSONB

is_active BOOLEAN

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

Indexes:

template_key

channel

461. NOTIFICATION_QUEUE TABLE

Columns:

id UUID PK

notification_id UUID FK

priority INTEGER

attempts INTEGER

next_retry_at TIMESTAMPTZ

status VARCHAR(50)

created_at TIMESTAMPTZ

Indexes:

status

next_retry_at

462. NOTIFICATION_PREFERENCES TABLE

Columns:

id UUID PK

user_id UUID FK

email_enabled BOOLEAN

sms_enabled BOOLEAN

whatsapp_enabled BOOLEAN

push_enabled BOOLEAN

marketing_enabled BOOLEAN

updated_at TIMESTAMPTZ

Indexes:

user_id UNIQUE

463. ANALYTICS_EVENTS TABLE

Universal Event Tracking

Columns:

id UUID PK

tenant_id UUID FK

user_id UUID FK NULL

event_name VARCHAR(255)

entity_type VARCHAR(100)

entity_id UUID

metadata JSONB

ip_address INET

created_at TIMESTAMPTZ

Indexes:

event_name

entity_type

entity_id

created_at

464. USER_ANALYTICS_DAILY TABLE

Columns:

id UUID PK

date DATE

new_users INTEGER

active_users INTEGER

verified_users INTEGER

premium_users INTEGER

created_at TIMESTAMPTZ

Unique:

(date)

465. SEARCH_ANALYTICS_DAILY TABLE

Columns:

id UUID PK

date DATE

total_searches INTEGER

ai_searches INTEGER

voice_searches INTEGER

property_searches INTEGER

project_searches INTEGER

created_at TIMESTAMPTZ

Unique:

(date)

466. CITY_ANALYTICS TABLE

Columns:

id UUID PK

city VARCHAR(150)

date DATE

property_views INTEGER

project_views INTEGER

leads_generated INTEGER

created_at TIMESTAMPTZ

Indexes:

city

date

467. CMS_PAGES TABLE

Columns:

id UUID PK

slug VARCHAR(255)

title VARCHAR(255)

content TEXT

status VARCHAR(50)

meta_title VARCHAR(255)

meta_description TEXT

published_at TIMESTAMPTZ

created_at TIMESTAMPTZ

updated_at TIMESTAMPTZ

Indexes:

slug UNIQUE

status

468. CMS_BLOGS TABLE

Columns:

id UUID PK

slug VARCHAR(255)

title VARCHAR(255)

excerpt TEXT

content TEXT

featured_image TEXT

author_id UUID FK

status VARCHAR(50)

published_at TIMESTAMPTZ

created_at TIMESTAMPTZ

Indexes:

slug UNIQUE

status

469. CMS_CATEGORIES TABLE

Columns:

id UUID PK

name VARCHAR(255)

slug VARCHAR(255)

created_at TIMESTAMPTZ

Indexes:

slug UNIQUE

470. CMS_TAGS TABLE

Columns:

id UUID PK

name VARCHAR(100)

slug VARCHAR(100)

created_at TIMESTAMPTZ

Indexes:

slug UNIQUE

471. CMS_FAQS TABLE

Columns:

id UUID PK

question TEXT

answer TEXT

category VARCHAR(255)

sort_order INTEGER

status BOOLEAN

created_at TIMESTAMPTZ

Indexes:

category

472. CMS_MENUS TABLE

Columns:

id UUID PK

menu_name VARCHAR(255)

location VARCHAR(100)

created_at TIMESTAMPTZ

Indexes:

location

473. CMS_MENU_ITEMS TABLE

Columns:

id UUID PK

menu_id UUID FK

title VARCHAR(255)

url TEXT

sort_order INTEGER
parent_id UUID NULL
created_at TIMESTAMPTZ

Indexes:

menu_id

474. SUPPORT_TICKETS TABLE

Columns:

id UUID PK
ticket_number VARCHAR(100)
user_id UUID FK
category VARCHAR(100)
priority VARCHAR(50)
subject VARCHAR(255)
description TEXT
status VARCHAR(50)
assigned_to UUID FK
created_at TIMESTAMPTZ
updated_at TIMESTAMPTZ

Indexes:

ticket_number UNIQUE
status
assigned_to

475. SUPPORT_MESSAGES TABLE

Columns:

id UUID PK
ticket_id UUID FK

sender_id UUID FK

message TEXT

attachment_url TEXT

created_at TIMESTAMPTZ

Indexes:

ticket_id

476. SUPPORT_CATEGORIES TABLE

Columns:

id UUID PK

name VARCHAR(255)

description TEXT

created_at TIMESTAMPTZ

Indexes:

name UNIQUE

477. SUPPORT_ESCALATIONS TABLE

Columns:

id UUID PK

ticket_id UUID FK

escalated_to UUID FK

reason TEXT

created_at TIMESTAMPTZ

Indexes:

ticket_id

478. KNOWLEDGE_BASE_ARTICLES TABLE

Columns:

id UUID PK

title VARCHAR(255)

slug VARCHAR(255)

content TEXT

category VARCHAR(255)

status VARCHAR(50)

created_at TIMESTAMPTZ

Indexes:

slug UNIQUE

479. FEATURE_FLAG_AUDIT TABLE

Columns:

id UUID PK

feature_flag_id UUID FK

old_value JSONB

new_value JSONB

changed_by UUID

created_at TIMESTAMPTZ

Indexes:

feature_flag_id

480. SYSTEM_JOBS TABLE

Columns:

id UUID PK

job_name VARCHAR(255)

job_type VARCHAR(100)

status VARCHAR(50)

payload JSONB

started_at TIMESTAMPTZ

completed_at TIMESTAMPTZ

created_at TIMESTAMPTZ

Indexes:

job_type

status

481. JOB_EXECUTION_HISTORY TABLE

Columns:

id UUID PK

job_id UUID FK

status VARCHAR(50)

execution_time_ms INTEGER

error_message TEXT

executed_at TIMESTAMPTZ

Indexes:

job_id

482. SYSTEM_HEALTH_LOGS TABLE

Columns:

id UUID PK

service_name VARCHAR(255)

status VARCHAR(50)

response_time_ms INTEGER

recorded_at TIMESTAMPTZ

Indexes:

service_name

recorded_at

483. FRAUD_FLAGS TABLE

Columns:

id UUID PK

entity_type VARCHAR(100)

entity_id UUID

fraud_type VARCHAR(100)

severity VARCHAR(50)

status VARCHAR(50)

created_at TIMESTAMPTZ

Indexes:

entity_type

entity_id

484. BLACKLISTED_ENTITIES TABLE

Columns:

id UUID PK

entity_type VARCHAR(100)

entity_value TEXT

reason TEXT

created_at TIMESTAMPTZ

Indexes:

entity_type

485. DATABASE FUNCTIONS

Mandatory Functions:

generate_slug()

calculate_subscription_usage()

update_search_index()
create_audit_log()
track_analytics_event()
create_notification()
assign_lead_round_robin()
calculate_lead_score()
expire_subscription()
generate_invoice()
update_inventory_summary()
create_price_history()
verify_geo_location()
detect_duplicate_property()
detect_duplicate_lead()

486. MATERIALIZED VIEWS

mv_property_statistics
mv_project_statistics
mv_revenue_statistics
mv_lead_statistics
mv_city_statistics
mv_builder_statistics
mv_agency_statistics

487. DATABASE VIEWS

vw_active_properties
vw_featured_properties
vw_active_projects
vw_subscription_summary

vw_payment_summary

vw_lead_summary

vw_user_summary

vw_builder_summary

vw_agency_summary

488. GLOBAL INDEX STRATEGY

B-Tree Indexes

GIN Indexes

PostGIS Indexes

Full Text Search Indexes

Composite Indexes

Partial Indexes

CRITICAL COMPOSITE INDEXES

properties:

(city, status)

(property_type, status)

(price, city)

projects:

(city, status)

(builder_id, status)

leads:

(status, assigned_to)

(source, created_at)

payments:

(status, paid_at)

(user_id, status)

489. GLOBAL RLS POLICIES

Public:

Approved Listings Only

Approved Projects Only

Public CMS Content

Buyer/Seller:

Own Data

Own Favorites

Own Visits

Own Messages

Broker:

Assigned Leads

Own Listings

Own Conversations

Builder:

Own Projects

Own Inventory

Own Leads

Agency:

Agency Users

Agency Leads

Agency Properties

Agency Projects

Admin:

Moderation Access

Super Admin:

Full System Access

490. DATABASE PERFORMANCE RULES

Query Timeout:

5 Seconds

Search Target:

<1 Second

Property Detail:

<500ms

Dashboard:

<2 Seconds

491. DATABASE BACKUP STRATEGY

Daily Backup

Weekly Backup

Monthly Backup

Yearly Archive

Point-In-Time Recovery

492. DATA RETENTION POLICY

Audit Logs:

7 Years

Payments:

10 Years

Invoices:

10 Years

Analytics:

3 Years

Notifications:

1 Year

Support Tickets:

5 Years

493. DATABASE SECURITY RULES

RLS Mandatory

Encrypted Secrets

Signed URLs

Audit Logging

Rate Limiting

IP Monitoring

Fraud Monitoring

494. FINAL DATABASE METRICS

Authentication Module:

18 Tables

Property Module:

25 Tables

Projects Module:

30 Tables

CRM Module:

35 Tables

Revenue Module:

28 Tables

Enterprise Module:

25 Tables

495. GRAND TOTAL DATABASE ARCHITECTURE

Estimated Tables:

161+

Views:

9+

Materialized Views:

7+

Functions:

15+

Triggers:

40+

Indexes:

300+

Policies:

100+

496. FINAL DATABASE STATUS

Authentication:

COMPLETE

RBAC:

COMPLETE

Properties:

COMPLETE

Projects:

COMPLETE

CRM:

COMPLETE

Messaging:
COMPLETE

Subscriptions:
COMPLETE

Payments:
COMPLETE

Advertisements:
COMPLETE

Analytics:
COMPLETE

CMS:
COMPLETE

Support:
COMPLETE

Security:
COMPLETE

Audit:
COMPLETE

RLS:
COMPLETE

Performance:
COMPLETE

Database Blueprint:
COMPLETE

PART 9F COMPLETE

DATABASE MASTER SCHEMA COMPLETE

PART 9 STATUS:
100% COMPLETE

TOTAL DATABASE TABLES:
161+

READY FOR:

- ✓ Supabase Implementation
- ✓ PostgreSQL Migration Creation
- ✓ Claude Code
- ✓ Cursor
- ✓ Windsurf
- ✓ Gemini
- ✓ Enterprise Development Team

END OF PART 9